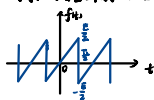


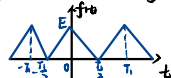
2. 周期锯齿脉冲信号



奇函数 $\Rightarrow a_n = 0$

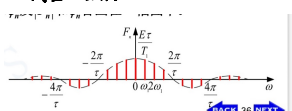
$$f(t) = \frac{E}{T} \sum_{n=-\infty}^{\infty} (-1)^n \frac{1}{n} \sin(n\pi t/T)$$

3. 周期三角脉冲信号



$$f(t) = \frac{E}{T} + \frac{4E}{T^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin^2(n\pi t/T) \cos(n\pi t/T)$$

二、傅里叶变换



$T \rightarrow \infty \Rightarrow$ 谱线间隔 $\rightarrow 0 \Rightarrow$ 周期信号 $\rightarrow$ 非周期信号
离散频谱 $\rightarrow$ 连续频谱

1. 频谱密度函数 (频谱函数)

$$F(\omega) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-j\omega t} dt$$

$$\Rightarrow F(\omega) T = 2\pi F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$T \rightarrow \infty$ 到 $\omega \rightarrow 0$

$$F(\omega) = \lim_{\omega \rightarrow 0} \frac{2\pi F(\omega)}{\omega} \text{ 为连续函数}$$

$$F(\omega) = \lim_{T \rightarrow \infty} F(\omega) T = \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} f(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

1) 傅里叶变换

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = |F(\omega)| e^{j\phi(\omega)}$$

若 $f(t)$ 为实函数

$$F(\omega) = \int_{-\infty}^{\infty} f(t) \cos(\omega t) dt - j \int_{-\infty}^{\infty} f(t) \sin(\omega t) dt$$

$$= \underbrace{R(\omega)}_{\text{偶}} + j \underbrace{X(\omega)}_{\text{奇}}$$

$$|F(\omega)| = \sqrt{R(\omega)^2 + X(\omega)^2} \quad |F(\omega)| \sim \omega \text{ 幅度谱 (带宽)}$$

$$\phi(\omega) = \arctan \frac{X(\omega)}{R(\omega)} \quad \phi(\omega) \sim \omega \text{ 相位谱}$$

1) 傅里叶逆变换

$$\text{傅里叶级数: } f(t) = \sum_{n=-\infty}^{\infty} F(\omega_n) e^{j\omega_n t} = \sum_{n=-\infty}^{\infty} \frac{F(\omega_n)}{\omega_n} e^{j\omega_n t} \cdot \omega_n$$

$$T \rightarrow \infty$$

$$F(\omega) = \lim_{\omega \rightarrow 0} \frac{2\pi F(\omega)}{\omega} \Rightarrow \frac{F(\omega)}{\omega} \rightarrow \frac{F(\omega)}{2\pi}$$

$$\Rightarrow f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

$$F(\omega) = F[f(t)] \quad f(t) = F^{-1}[F(\omega)]$$

2. 傅里叶变换物理意义

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(\omega)| e^{j(\omega t + \phi(\omega))} d\omega$$

$$\text{实函数} \Rightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(\omega)| \cos(\omega t + \phi(\omega)) d\omega$$

$$= \frac{1}{\pi} \int_0^{\infty} |F(\omega)| \cos(\omega t + \phi(\omega)) d\omega$$

无穷多振幅小于 $|F(\omega)| d\omega$ 的连续正弦信号之和

3. 傅里叶变换存在的条件

充分条件: $f(t)$ 在有限区间内绝对可积, 即 $\int_{-\infty}^{\infty} |f(t)| dt < \infty$

四. 典型非周期函数的傅里叶变换

1. 单边指数信号

$$f(t) = \begin{cases} e^{-at}, & t \geq 0 \\ 0, & t < 0 \end{cases}, a \text{ 实数}$$

$$\text{则 } F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = \frac{1}{a + j\omega}$$



2. 双边指数信号

$$f(t) = e^{-a|t|}, a > 0, \omega \text{ 实数}$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = \frac{2a}{a^2 + \omega^2}$$



3. 矩形脉冲信号

$$f(t) = E \left[e^{j\omega_1 t} - e^{j\omega_2 t} \right]$$

$$\text{则 } F(\omega) = E \tau \text{Sa} \left(\frac{\omega \tau}{2} \right)$$

$$G_T(\omega) \leftrightarrow \tau \text{Sa} \left(\frac{\omega \tau}{2} \right) \quad (E=1)$$

$$2\pi G_T(\omega) \leftrightarrow \tau \text{Sa} \left(\frac{\omega \tau}{2} \right)$$

不满足绝对可积条件的信号

4. 直流信号 $f(t) = E$ ($E \leftrightarrow 2\pi \delta(\omega)$)

$f(t)$ 看作矩形脉冲信号 ($T \rightarrow \infty$)

$$F(\omega) = \lim_{T \rightarrow \infty} E T \text{Sa} \left(\frac{\omega T}{2} \right) = \lim_{T \rightarrow \infty} 2\pi E \frac{T}{\omega} \text{Sa} \left(\frac{\omega T}{2} \right)$$

$$\left(\begin{aligned} \delta(\omega) &= \lim_{T \rightarrow \infty} \frac{T}{\omega} \text{Sa} \left(\frac{\omega T}{2} \right) \\ F(\omega) &= 2\pi E \delta(\omega) \end{aligned} \right)$$

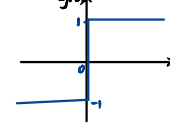
$$\int_{-\infty}^{\infty} \text{Sa} \left(\frac{\omega T}{2} \right) d\omega = \pi$$

$$\int_{-\infty}^{\infty} \text{Sa} \left(\frac{\omega T}{2} \right) d\omega = \frac{2\pi}{T}$$

$$\lim_{T \rightarrow \infty} \text{Sa} \left(\frac{\omega T}{2} \right) = \frac{2\pi}{\omega} \delta(\omega)$$

5. 符号信号 $\text{sgn}(t) \leftrightarrow \frac{1}{j\omega}$

$$f(t) = \text{sgn}(t) = \begin{cases} 1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$



$$f(t) = \begin{cases} e^{-at}, & t > 0 \\ -e^{-at}, & t < 0 \end{cases}, a > 0$$

$$F(\omega) = \frac{-j2a}{a^2 + \omega^2}$$

$$F(\omega) = \lim_{a \rightarrow 0} F(\omega) = \frac{1}{j\omega}$$

$$|F(\omega)| = \frac{1}{|\omega|}$$

$$\phi(\omega) = \begin{cases} -\frac{\pi}{2}, & \omega > 0 \\ \frac{\pi}{2}, & \omega < 0 \end{cases}$$

五. 冲激函数和阶跃函数的傅里叶变换

1. 冲激函数的傅里叶变换

$$1) F[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

$$\delta(t) \leftrightarrow 1$$

$$12) F^{-1}[\delta(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega = \frac{1}{2\pi}$$

$$\delta(t) \leftrightarrow 1 \rightarrow F^{-1}[1] = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega t} d\omega = \delta(t)$$

$$1 \leftrightarrow 2\pi \delta(\omega) \rightarrow F[1] = \int_{-\infty}^{\infty} e^{-j\omega t} dt = 2\pi \delta(\omega)$$

$$\Rightarrow \int_{-\infty}^{\infty} e^{j\omega t} dt = 2\pi \delta(\omega)$$

$$\delta^{(n)}(t) \leftrightarrow (j\omega)^n$$

2. 冲激函数偶的傅里叶变换

$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega t} d\omega$$

$$\delta'(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} j\omega e^{j\omega t} d\omega$$

$$F[\delta'(t)] = j\omega$$

$$\delta(\omega) = \delta(-\omega) \Rightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega t} d\omega = \delta(t)$$

$$2\pi \delta(\omega) = \int_{-\infty}^{\infty} e^{-j\omega t} dt = 2\pi \delta(\omega)$$

$$F[1] = j\omega \delta(\omega)$$

3. 连续函数的傅里叶变换

$$W(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$\Rightarrow F[f(t)u(t)] = \pi \delta(\omega) + \frac{j}{\omega}$$

六. 傅里叶变换的基本性质

1. 对称性 (易性)

若 $f(t) \leftrightarrow F(\omega)$ 则 $F(t) \leftrightarrow 2\pi f(-\omega)$

$f(t)$ 为偶 $\Rightarrow F(\omega)$ 为偶

$f(t)$ 为奇 $\Rightarrow F(\omega)$ 为奇

信号的时域波形与信号的频谱图形有互相关关系

$$\delta(t) \leftrightarrow 1 \quad \text{则} \quad F[\delta(t)] = 1 \leftrightarrow 2\pi \delta(\omega)$$

$$\text{Sgn}(t) \leftrightarrow \frac{1}{j\omega} \quad \text{则} \quad F[\text{Sgn}(t)] = \frac{1}{j\omega} \leftrightarrow 2\pi \text{Sgn}(\omega)$$

$$\Rightarrow \frac{1}{j\omega} \leftrightarrow -j\pi \text{Sgn}(\omega)$$

证明: $f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$

$$F(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$2\pi f(-\omega) = \int_{-\infty}^{\infty} F(t) e^{j\omega t} dt$$

$$F(t) \leftrightarrow 2\pi f(-\omega)$$

矩形脉冲的频谱为 sinc 函数

Sa 形脉冲的频谱为矩形函数 $2\pi \text{Sa}(\omega) \leftrightarrow \tau \text{Sa}(\frac{\tau}{2})$

2. 线性

若 $f_1(t) \leftrightarrow F_1(\omega)$, $f_2(t) \leftrightarrow F_2(\omega)$ 则 $a f_1(t) + b f_2(t) \leftrightarrow a F_1(\omega) + b F_2(\omega)$

3. 奇偶虚实性

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} f(t) \cos \omega t dt - j \int_{-\infty}^{\infty} f(t) \sin \omega t dt$$

$$F(\omega) = R(\omega) + jX(\omega)$$

$$|F(\omega)| = \sqrt{R^2(\omega) + X^2(\omega)} \quad f(t) = a \cos \omega t \left[\frac{2\pi}{\omega} \right]$$

1) $f(t)$ 是实函数 $F(\omega)$ 偶 $X(\omega)$ 奇

$\Rightarrow |F(\omega)|$ 偶 $\phi(\omega)$ 奇

$f(t)$ 实偶 $\Rightarrow F(\omega)$ 实偶

$f(t)$ 实奇 $\Rightarrow F(\omega)$ 虚奇

2) $f(t)$ 是虚函数

$$R(\omega) = -R(-\omega) \quad X(\omega) = X(-\omega)$$

$$f^*(t) \leftrightarrow F^*(\omega) \quad \text{则} \quad f(-t) \leftrightarrow F^*(\omega)$$

$$f^*(t) \leftrightarrow F^*(\omega) \quad f^*(t) \leftrightarrow F^*(\omega)$$

4. 尺度变换特性

若 $f(t) \leftrightarrow F(\omega)$, 则 $f(at) \leftrightarrow \frac{1}{|a|} F(\frac{\omega}{a})$, a 为实常数

$$F(\omega) = \int_{-\infty}^{\infty} f(t) dt \quad \text{则} \quad F(\omega)$$
 所覆盖的面积等于 $F(0)$

$$f(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) d\omega$$

$$\left. \begin{aligned} f(0) &= F(0) \\ f(0) &= 2\pi f(0) \end{aligned} \right\} \Rightarrow B_T = 2\pi \quad (T \text{ 为 } f(t) \text{ 等效带宽})$$

若 $f(t) \leftrightarrow F(\omega)$, 则 $f(t-t_0) \leftrightarrow F(\omega) e^{-j\omega t_0}$

$f(t+t_0) \leftrightarrow F(\omega) e^{j\omega t_0}$

6. 频移特性:

若 $f(t) \leftrightarrow F(\omega)$, 则 $f(t) e^{j\omega_0 t} \leftrightarrow F(\omega - \omega_0)$

$f(t) e^{-j\omega_0 t} \leftrightarrow F(\omega + \omega_0)$

$$\left\{ \begin{aligned} f(t) \cos(\omega_0 t) &\leftrightarrow \frac{1}{2} [F(\omega - \omega_0) + F(\omega + \omega_0)] \\ f(t) \sin(\omega_0 t) &\leftrightarrow \frac{j}{2} [F(\omega + \omega_0) - F(\omega - \omega_0)] \end{aligned} \right.$$

$$\left\{ \begin{aligned} \cos(\omega_0 t) &\leftrightarrow \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] \\ \sin(\omega_0 t) &\leftrightarrow j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)] \end{aligned} \right.$$

$$1 \leftrightarrow 2\pi \delta(\omega)$$

$$\cos(\omega_0 t) \leftrightarrow \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

7. 卷积特性

若 $f(t) \leftrightarrow F(\omega)$ 则 $f(t) \leftrightarrow j\omega F(\omega)$

$1-j\omega f(t) \leftrightarrow F(\omega)$

$1-j\omega f(t) \leftrightarrow F(\omega)$

8. 积分特性:

若 $f(t) \leftrightarrow F(\omega)$ 则 $\int_{-\infty}^t f(\tau) d\tau \leftrightarrow \frac{F(\omega)}{j\omega} + \pi F(\omega) \delta(\omega)$

$-\frac{F(\omega)}{j\omega} + \pi F(\omega) \delta(\omega) \leftrightarrow \int_{-\infty}^t f(\tau) d\tau$

若 $F(0) \neq 0$ 时 $\int_{-\infty}^t f(\tau) d\tau \leftrightarrow \frac{F(\omega)}{j\omega}$

七. 卷积

卷积定理:

① 若: $f_1(t) \leftrightarrow F_1(\omega)$, $f_2(t) \leftrightarrow F_2(\omega)$ 则 $f_1(t) * f_2(t) \leftrightarrow F_1(\omega) F_2(\omega)$

② 若: $f_1(t) \leftrightarrow F_1(\omega)$, $f_2(t) \leftrightarrow F_2(\omega)$ 则 $f_1(t) f_2(t) \leftrightarrow \frac{1}{2\pi} F_1(\omega) * F_2(\omega)$

八. 周期信号的傅里叶变换

1. 正弦信号的傅里叶变换

$$\cos(\omega_0 t) \leftrightarrow \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$\sin(\omega_0 t) \leftrightarrow j\pi [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

$$1 \leftrightarrow 2\pi \delta(\omega)$$

$$F[e^{j\omega_0 t}] = 2\pi \delta(\omega - \omega_0)$$

$$F[e^{-j\omega_0 t}] = 2\pi \delta(\omega + \omega_0)$$

2. 一般周期信号的傅里叶变换

周期为 T , $\omega_0 = 2\pi/T$

$$f(t) = \sum_{n=-\infty}^{\infty} F_n(\omega_0) e^{jn\omega_0 t} \quad \text{其中} \quad F_n(\omega_0) = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt$$

$$F[f(t)] = F\left[\sum_{n=-\infty}^{\infty} F_n(\omega_0) e^{jn\omega_0 t}\right] = \sum_{n=-\infty}^{\infty} F_n(\omega_0) F[e^{jn\omega_0 t}]$$

$$= 2\pi \sum_{n=-\infty}^{\infty} F_n(\omega_0) \delta(\omega - n\omega_0)$$

由一些冲激函数组成

Eq: $\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$ $\delta_T(t) \leftrightarrow 1$

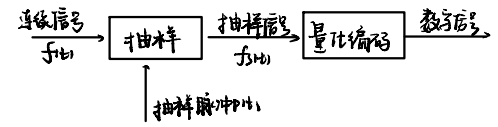
$\delta_T(t)$ 的傅里叶级数谱系数: $F_n(\omega_0) = \frac{1}{T}$

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} F_n(\omega_0) e^{jn\omega_0 t} = \frac{1}{T} \sum_{n=-\infty}^{\infty} e^{jn\omega_0 t}$$

$$\Rightarrow F(\omega) = \frac{1}{T} \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \delta(\omega - n\omega_0) = \omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$$

九. 抽样信号的傅里叶变换

1. 抽样



2. 时域抽样

抽样脉冲序列及其傅里叶变换 抽样周期 T_s , $\omega_s = \frac{2\pi}{T_s}$

$$p(t) \leftrightarrow p(\omega) = 2\pi \sum_{n=-\infty}^{\infty} p_n \delta(\omega - n\omega_s)$$

① 周期矩形脉冲序列 $p_n = p_n(\omega_s) = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} p(t) e^{-jn\omega_s t} dt$

② 周期单位冲激序列

时域抽样 $f_s(t) = f(t) p(t) \leftrightarrow F_s(\omega) = \frac{1}{T_s} F(\omega) * p(\omega)$

$$= \sum_{n=-\infty}^{\infty} p_n F(\omega - n\omega_s)$$